| Ex. No.:2 | AI-Powered Blog Content Generator | URK23CS1167 |
|---|---|---|
| 13.08.2026 | | |

## Aim:

To build an AI-powered Blog Content Generator using Ollama that generates blogs, titles, summaries, SEO keywords, and meta descriptions based on user prompts.

## Problem Statement:

Design and develop a modular AI-powered Blog Content Generator using Ollama. The application should generate high-quality blogs from user-provided topics, writing styles, and word counts. It should also generate blog titles, summaries, SEO keywords, and meta descriptions while intelligently routing user requests using an LLM-based intent detector.

## Software Requirements:

| Software | Version |
|---|---|
| Python | 3.13.5 |
| Ollama | 0.40.13 |
| VS Code | 1.144.0 |
| Pandas | 3.1.0 |
| Sentence Transformers | 6.2.0 |
| Flask | 3.1.3 |
| LLM Model | Llama 3.2 (3B) via Ollama |

## Input Parameters:

| Parameters | Description |
|---|---|
| Blog Topic | Subject of the blog. Example: Electric Vehicles in India |
| Writing Style | Professional, Informative, Conversational, Storytelling, Technical or Listicle |
| Tone | Friendly, Formal, Persuasive, Inspirational, Neutral or Humorous |
| Word count | Target length of the blog (150 – 1500 words). Example: 400 |
| Target Audience | Who the blog is written for. Example: Engineering students |
| Additional Instructions | Optional extra guidance. Example: Include benefits, challenges and future |

## System Architecture:

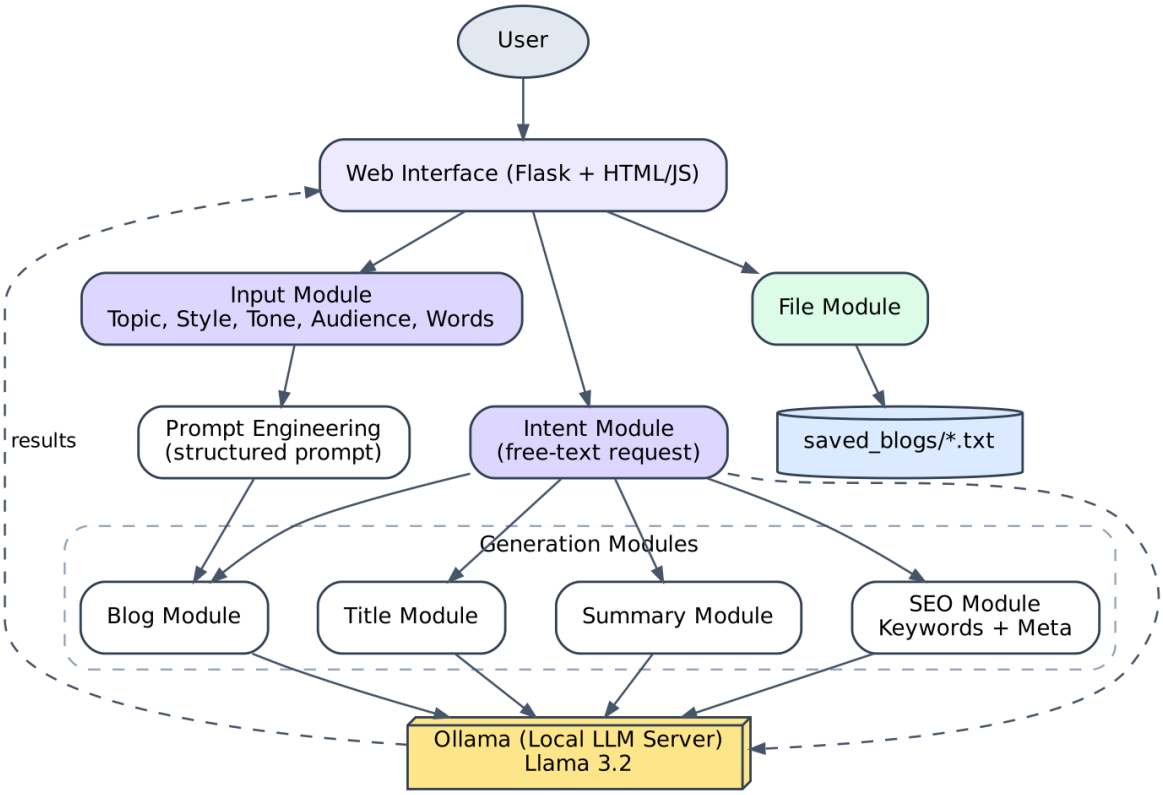



*Fig. 2.1 System architecture*

## Module Design:

| Module | Responsibility |
|---|---|
| Input Module | Collects and validates topic, writing style, tone, audience, word count and additional instructions from the web form. |
| Blog Module | Builds a structured prompt (prompt engineering) and generates the blog with title, Introduction, main sections and Conclusion using Llama 3.2. |
| Title Module | Generates five catchy, SEO-friendly blog titles under 70 characters. |
| Summary Module | Generates a concise summary of about 60 words for the generated blog. |
| SEO Module | Extracts 8 SEO keywords (short and long-tail) and writes a meta description within 160 characters with a search-result preview. |
| Intent Module | Understands a free-text request (e.g. 'suggest catchy headings') and routes it to the Blog, Title, Summary, SEO, Meta or Save action. |
| File Module | Saves the blog with its titles, summary, keywords and meta description to a .txt file in the saved_blogs folder. |

## Implementation:

• GitHub Link - https://github.com/MONICK10/GPT4-Exp2-Blog-Content-Generator

• LinkedIn Post Link - <add LinkedIn post link in config.json>
• Run: `python app.py` → `http://localhost:5002`

## Output Screenshots:

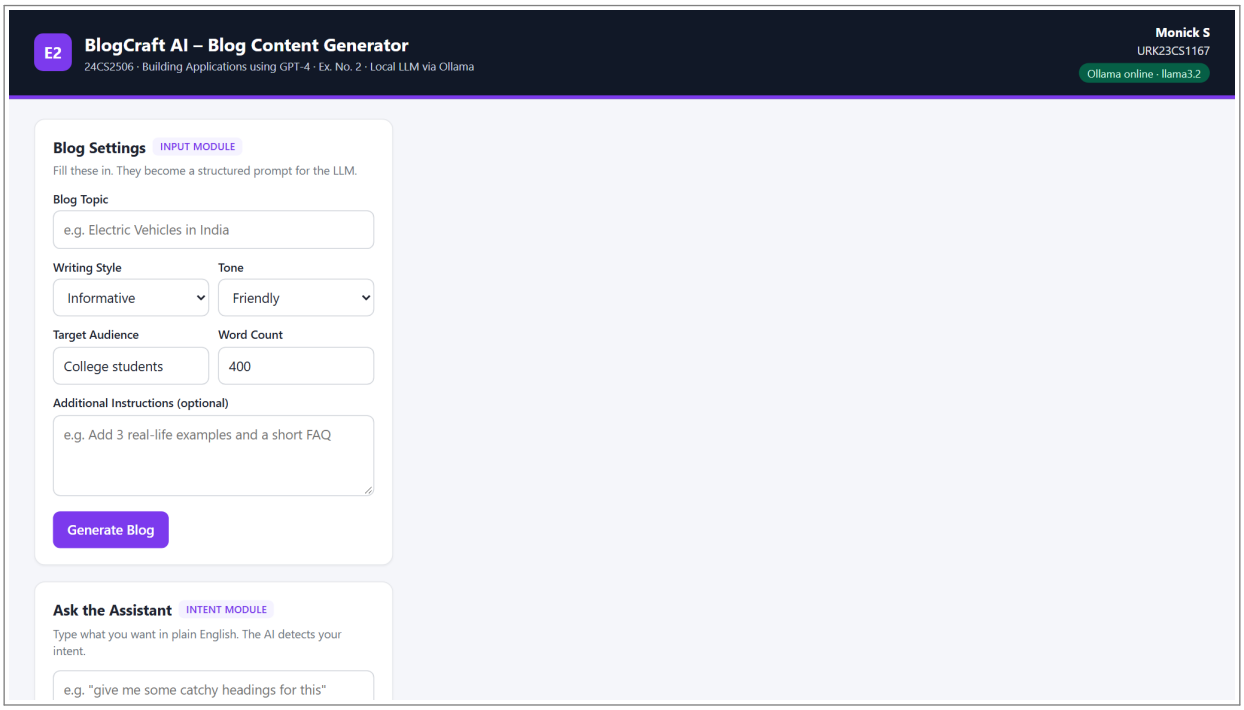



*Fig. 2.2 Home Screen*

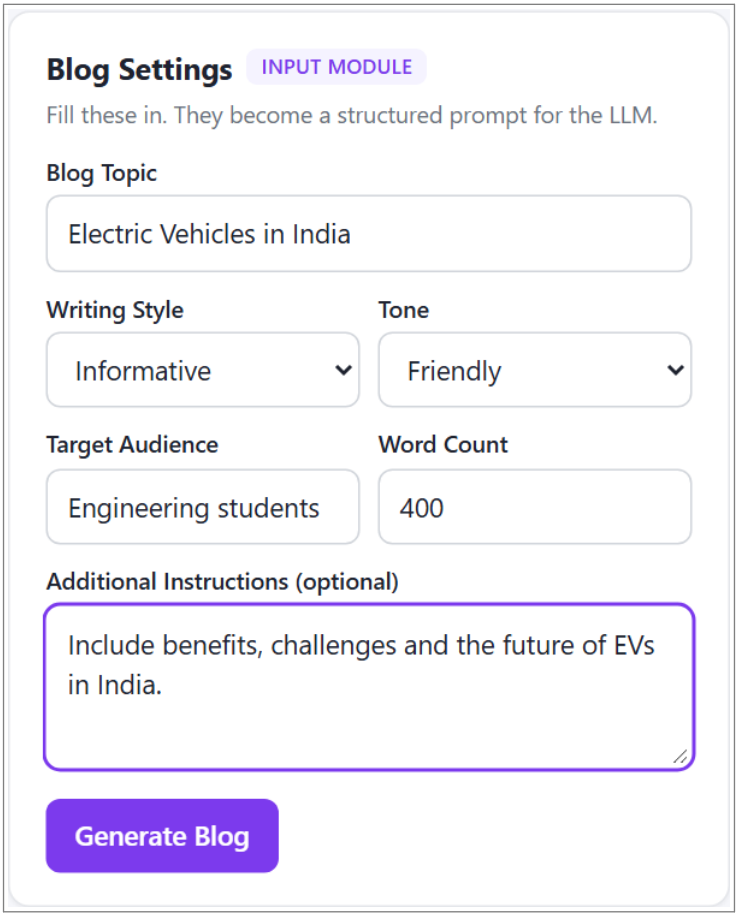



*Fig. 2.3 User Input*

**Generated Blog** BLOG MODULE 437 words

Generate Titles | Summary | SEO Keywords | Meta Description | Save Blog | Copy

**Electric Vehicles in India**

As engineering students, you're likely familiar with the concept of sustainable energy and its impact on the environment. One of the most promising ways to reduce our carbon footprint is by transitioning to electric vehicles (EVs). In this blog, we'll delve into the world of electric vehicles in India, exploring their benefits, challenges, and the future of EVs in our country.

**Introduction**

India has set ambitious targets to reduce its reliance on fossil fuels and combat climate change. Electric vehicles, with their zero-emission technology, are an attractive solution. The Indian government has announced plans to promote EV adoption, making it an exciting time for the industry. In this blog, we'll examine the current state of electric vehicles in India and what the future holds.

**Benefits of Electric Vehicles**

Electric vehicles offer several benefits, including:

- Zero Tailpipe Emissions: EVs produce no emissions, reducing air pollution and greenhouse gas emissions.
- Lower Operating Costs: EVs are cheaper to run, with lower fuel costs (electricity is generally cheaper than petrol or diesel).
- Smooth and Quiet Ride: EVs have a smoother and quieter ride due to their electric motor.
- Government Incentives: The Indian government offers various incentives, such as tax exemptions and subsidies, to encourage EV adoption.

**Challenges Facing Electric Vehicles in India**

While EVs offer many benefits, there are several challenges that need to be addressed:

- Range Anxiety: The limited range of EVs can make long-distance travel challenging.
- Charging Infrastructure: The lack of charging stations and slow charging speeds can be a major barrier to adoption.
- High Upfront Costs: EVs are often more expensive than their petrol or diesel counterparts

*Fig. 2.4 Blog Generation*

**Engineered Prompt** PROMPT ENGINEERING

```
Write a 400-word informative blog on "Electric Vehicles in India".
Target audience: Engineering students.
Tone: Friendly.
Structure:
# <Blog Title>
## Introduction
## <2-4 main section headings>
## Conclusion
Use simple English, short paragraphs and bullet points where useful.
Additional instructions: Include benefits, challenges and the future of EVs in India.
```

*Fig. 2.5 Prompt Engineering*

**Blog Titles** TITLE MODULE

1. **Revolutionizing India's Roads: The Future of EVs**
2. **Go Electric: Unlocking India's EV Potential**
3. **What's Holding Back EV Adoption in India?**
4. **India's EV Revolution: Challenges and Opportunities**
5. **Will Electric Vehicles Make India Greener?**

*Fig. 2.6 Blog Title Generation*

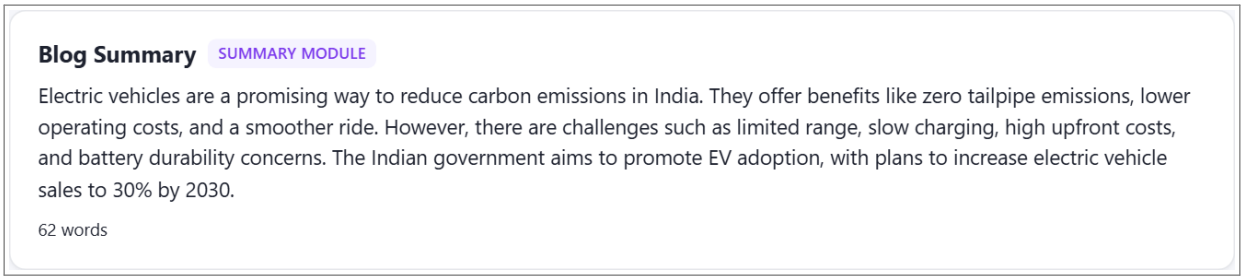



*Fig. 2.7 Summary Generation*

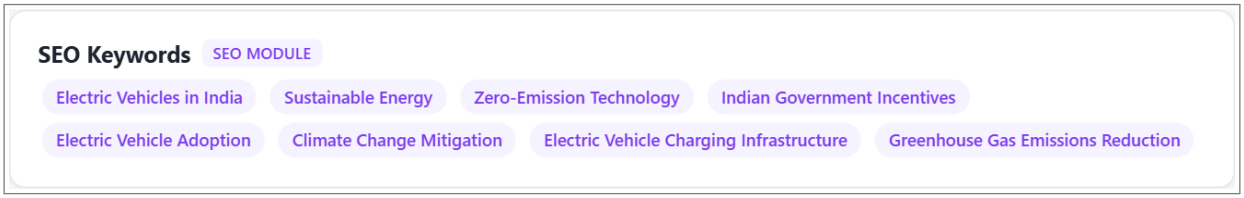



*Fig. 2.8 SEO Keywords*

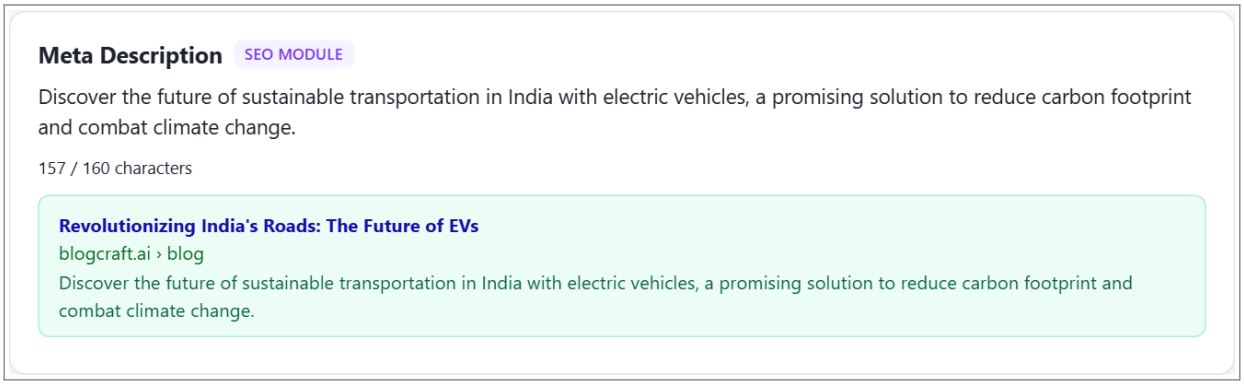



*Fig. 2.9 Meta Description*




*Fig. 2.10 Blog Saved Successfully*

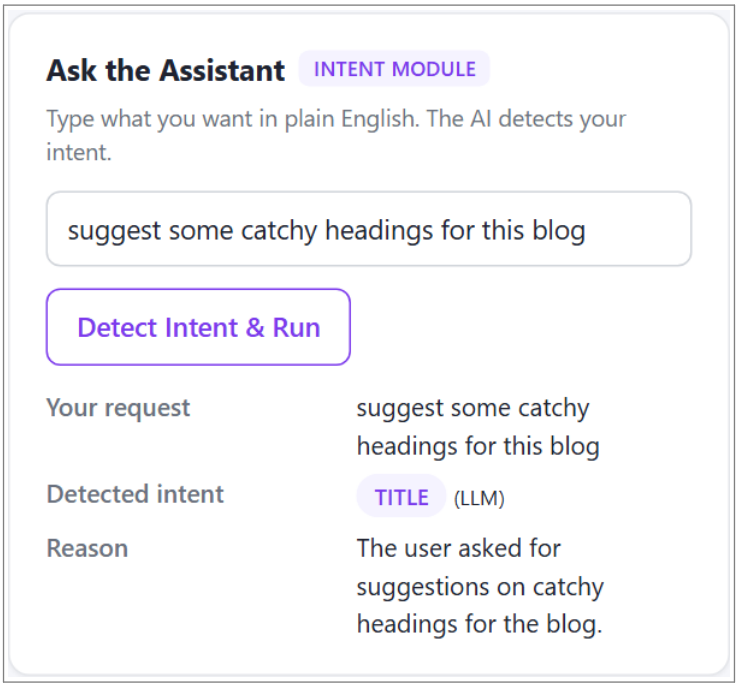



*Fig. 2.11 Intent Detection*

**Result:**

Thus, an AI-powered Blog Content Generator was successfully developed using Ollama. The application generates blogs based on user prompts, creates titles, summaries, SEO keywords, and meta descriptions, saves the generated content to a text file, and demonstrates the use of local Large Language Models (LLMs) for content generation without requiring cloud-based APIs.